What is a model?

Computational Methods for Macroeconomics — Session 1

Vasudeva Ramaswamy · CIDE · Fall 2026

A model is a set of equations.

Solving a model means finding the values of its unknowns.

What is a model?

In economics, a **model** is a set of equations.

$$\alpha_1 x + \beta_1 y = \gamma_1$$
$$\alpha_2 x + \beta_2 y = \gamma_2$$

This is a model. A small one, but a model.

Unknowns and parameters

$$\alpha_1 x + \beta_1 y = \gamma_1$$
$$\alpha_2 x + \beta_2 y = \gamma_2$$

- The **unknowns** are x and y. To **solve the model** is to find their values.
- The **parameters** α, β, γ are values that are *known*.

So if $\alpha_1 = 1$, $\beta_1 = 3$, $\gamma_1 = 7$, $\alpha_2 = 4$, $\beta_2 = 5$, $\gamma_2 = 14$, we would *actually* solve

$$x + 3y = 7, \qquad 4x + 5y = 14 \qquad \Longrightarrow \qquad x = 1,\ y = 2.$$

Colour code for the whole course: parameters are known numbers, unknowns are what we solve for.

When is a model “solvable”?

These are **linear** equations: each unknown appears independently of other unknowns — no squares, no products of unknowns, no functions like log or exp.

- Rule of thumb: for linear systems you generally need as many *independent* equations as unknowns.

Here: 2 unknowns, 2 equations — good.

“Solvable” (in this course)

1. A solution *exists* (existence)
2. There is only *one* solution (uniqueness)

When either fails, we must *add or remove conditions* to restore it.

Models with a time element

Many models in economics are **dynamic**: their equations have a time element.

$$\alpha_1 x_t + \beta_1 y_{t+1} + \gamma_1 y_t = 0$$
$$\alpha_2 x_{t+1} + \beta_2 y_t + \gamma_2 x_t = 0$$

$$\text{for } t = 0, 1, 2$$

Same parameters as before — but now every variable carries a **time subscript**: x_t is the value of x in period t.

What the subscript really says

"For $t = 0, 1, 2$" means these two lines are really **three sets of equations**:

$t = 0$	$t = 1$	$t = 2$
$\alpha_1 x_0 + \beta_1 y_1 + \gamma_1 y_0 = 0$	$\alpha_1 x_1 + \beta_1 y_2 + \gamma_1 y_1 = 0$	$\alpha_1 x_2 + \beta_1 y_3 + \gamma_1 y_2 = 0$
$\alpha_2 x_1 + \beta_2 y_0 + \gamma_2 x_0 = 0$	$\alpha_2 x_2 + \beta_2 y_1 + \gamma_2 x_1 = 0$	$\alpha_2 x_3 + \beta_2 y_2 + \gamma_2 x_2 = 0$

Writing all six out is tedious, so we write them *compactly* using t:

$$\alpha_1 x_t + \beta_1 y_{t+1} + \gamma_1 y_t = 0, \qquad \alpha_2 x_{t+1} + \beta_2 y_t + \gamma_2 x_t = 0.$$

The "D" in DSGE

This course is about such **dynamic** systems — equations that link today's values to tomorrow's.

Counting equations and unknowns

BOARD / DISCUSSION

What are the unknowns here? And how many equations do we have?

- The unknowns: x_0, x_1, x_2 and $y_0, y_1, y_2 \ldots$ but look at the $t = 2$ equations: x_3 and y_3 appear too. They are unknown as well.
- So: **6 equations, 8 unknowns.** By our definition, this model is *not solvable* — unless two values are given to us.
- With T finite, almost *any* two values will do.

What if time never ends? $T \to \infty$

Infinitely many equations, infinitely many unknowns. Can it still be slved?

- One simple case: if x_0 and y_0 are given, rearrange the period-t equations (assuming $\beta_1 \neq 0$, $\alpha_2 \neq 0$):

$$y_{t+1} = -\frac{\alpha_1 x_t + \gamma_1 y_t}{\beta_1}, \qquad x_{t+1} = -\frac{\gamma_2 x_t + \beta_2 y_t}{\alpha_2}.$$

  Each period's values give you the next period's. Starting from (x_0, y_0) you can roll the whole future forward, one period at a time. This is an **initial value problem**.
- But what if x_0 and/or y_0 are *not* given? Is it still solvable?

A preview

DSGE models run into exactly this situation all the time — and under the right conditions they can *still* be solved. **Blanchard–Kahn** conditions: what those are, and why they work, comes later in this course.

A third variable

$$\alpha_1 x_t + \beta_1 y_{t+1} + \gamma_1 y_t + \delta_1 z_t = 0$$
$$\alpha_2 x_{t+1} + \beta_2 y_t + \gamma_2 x_t + \delta_2 z_t = 0$$

$$\text{for } t = 0, 1, 2$$

z is a *variable* — its value changes over time, hence the subscript.

- Can we solve for x, y *and* z, even knowing x_0 and y_0? **No**: we added three unknowns (z_0, z_1, z_2) but zero new equations.
- But if the *entire path* of z is known — z_0, z_1, z_2 all given — we have no trouble at all.

Endogenous and exogenous variables

Exogenous variables (Spanish: *variables exógenas*)

Variables like z, whose whole sequence of values is known from *outside* the model. They are like parameters whose values change every period in a known way.

Endogenous variables (Spanish: *variables endógenas*)

Variables like x and y, whose values are determined *inside* the model — you have to solve for them, and each one depends on all the other unknowns in the system.

Colour code, updated: parameters · endogenous · exogenous.

Vector and matrix form

Gather the endogenous variables into a vector, $s_t = \begin{bmatrix} x_t \\ y_t \end{bmatrix}$. Linear dynamic systems can then always be written as

$$A\, s_{t+1} = B\, s_t + C\,.$$

For our two-equation system (second equation first):

$$\underbrace{\begin{bmatrix} \alpha_2 & 0 \\ 0 & \beta_1 \end{bmatrix}}_{A} \begin{bmatrix} x_{t+1} \\ y_{t+1} \end{bmatrix} = \underbrace{\begin{bmatrix} -\gamma_2 & -\beta_2 \\ -\alpha_1 & -\gamma_1 \end{bmatrix}}_{B} \begin{bmatrix} x_t \\ y_t \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ 0 \end{bmatrix}}_{C}$$

Why is this better?

- Because **linear algebra** is a vast, well-studied field — writing models this way gives us many useful tools.
- Because linear systems with any number of lags and leads can be rewritten in the above first-order form.

Life is not linear

Often the equations are **nonlinear**:

$$\alpha_1 \log(1 + x_t) = y_{t+1}^2 \, z_{t+1}, \qquad \alpha_2 x_{t+1} = \frac{1}{1 - y_t + \gamma_2 z_{t+1}}.$$

In general form, with $s_t = [x_t, y_t]'$:

$$F(s_{t+1},\, s_t,\, z_{t+1},\, z_t) = 0.$$

- How do we solve *this*? In general: **not with pen and paper.**
- This course covers the ways that work: numerical methods, exact and approximate solutions.
- And even then, we will constantly rely on results from linear algebra — which is why we introduced the linear case first.

When the future is uncertain

So far, the exogenous z had a *known* path. In real-world economics, it usually doesn't. Instead, z is **stochastic**: a random variable whose *general behaviour* is known, but whose exact value each period is not.

Today's z_t we observe; tomorrow's z_{t+1} we don't. And since tomorrow's endogenous variables depend on tomorrow's z, they are uncertain too. So we write the system with an **expectations operator** on the future:

$$A\,\mathbb{E}_t[s_{t+1}] = B\,s_t + C z_t.$$

The "S" in DSGE

Stochastic: the outside forces hitting the model are random, so the model's equations involve *expectations* of the future.

The expectations operator

Standing in period t, our *expectation* of tomorrow's z is its probability-weighted average:

$$\mathbb{E}_t[z_{t+1}] = \sum_{i \in I} z_i \, \Omega_i ,$$

where I indexes all possible values of z (its "state space") and Ω_i is the probability that $z_{t+1} = z_i$.

Example

Suppose tomorrow z takes the value 1, 2, or 3 with probabilities 30%, 20%, 50%. Then

$$\mathbb{E}_t[z_{t+1}] = 1(0.3) + 2(0.2) + 3(0.5) = 2.2.$$

Tomorrow, z may actually turn out to be 1. But standing in period t, we don't know that — 2.2 is our best guess.

Whose expectations?

Stochasticity matters — and it makes models harder to solve. It also opens a deeper question: **how are expectations formed?**

- In an economy, expectations are formed by *agents*: households, firms, the government.
- When I say "I expect it to rain tomorrow," I did *not* compute a probability-weighted average of rainfall outcomes.
- So how do we align the expectations agents actually form with the mathematical operator $\mathbb{E}_t$?

The literature offers several answers. We will review some, and focus on the most important one: **Rational Expectations** — defined and used extensively in this course.

Where do the equations come from?

So far the equations fell from the sky. In economics, they don't.

- You have met models like the RBC and New Keynesian models, and models of specific behaviour — consumption, investment.
- In DSGE models, the equations describe the **behaviour of all the agents in an economy** as they *interact* with one another.
- So a change in behaviour by *any* agent affects the *entire* economy.

Everything affects everything else

Suppose households decide to supply more labor.

$$\text{more labor supplied} \Rightarrow \text{wages fall} \Rightarrow \text{firm profits rise}\ldots$$
$$\ldots\text{but household incomes fall} \Rightarrow \text{demand for output falls} \Rightarrow \text{profits fall}$$
$$\text{lower incomes} \Rightarrow \text{less saving} \Rightarrow \text{interest rates rise} \Rightarrow \text{borrowing is costlier} \Rightarrow \text{output falls}$$
$$\Rightarrow \ldots$$

The same initial change hit the firm both positively *and* negatively, and the cycle seems endless — everything affects everything else.

The "GE" in DSGE

Under specific conditions, the cycle *does* settle down. That is **General Equilibrium** — and DSGE models let us predict how a single change plays out across the whole economy, over time.

Putting the letters together

D	**Dynamic** — variables carry time subscripts; equations link today to tomorrow.
S	**Stochastic** — exogenous forces are random; expectations of the future enter the equations.
GE	**General Equilibrium** — the equations come from all agents' behaviour, interacting at once.

DSGE models are also just a set of equations. Solving it means finding the unknowns, and "solvable" means a solution *exists* and is *unique*.

Course Overview

		Sessions
Part I	How to solve these systems	2–8
Part II	Building real models (Dynare & Helix)	9–14
Part III	Taking models to data: estimation	15–20
Part IV	Many different agents: heterogeneity	21–27
Closing	Your project presentations	28

Every tool is introduced gently, used hands-on, and reused later. By the end you will build, solve, and estimate DSGE models yourself.

Before next session: software

- **Dynare** — our workhorse for building and estimating models. Free; runs on MATLAB or on free GNU Octave.
- **Helix** — browser-based solver we will use later in the course. Nothing to install.
- Installation links and step-by-step instructions are on the course page. Please have Dynare running before next week — and write to me if you get stuck.

BOARD / DISCUSSION

Quick show of hands: who already has MATLAB? Octave? Neither?

Next class: we start solving with pen-and-paper